

MD. SHAJIBUL ISLAM

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🌐 ShajibEwuCse19 🔄 ShajibEwuCse19 🏠 Google Scholar 📁 Portfolio

ACADEMIC CREDENTIALS

East West University January 2019 – September 2023
Bachelor of Science in Computer Science & Engineering CGPA: 3.60/4.00
Major: Intelligent Systems and Data Science

PROFESSIONAL EXPERIENCE

Pathao Limited Dhaka, Bangladesh
Software Engineer I January 2024 – Present

- ❖ Contributed as a software engineer at Pathao, a leading consumer technology platform in Bangladesh providing large-scale digital services across mobility, logistics, food delivery, e-commerce, and financial technology.
- ❖ Designed, developed, and maintained production-level software systems used by a large user base, with an emphasis on reliability, scalability, code quality, and long-term maintainability.
- ❖ Collaborated with cross-functional engineering, product, and design teams to translate real-world user and business requirements into robust software solutions in a fast-paced technology environment.

East West University Dhaka, Bangladesh
Graduate Teaching Assistant January 2023 – December 2023

- ❖ Assisted undergraduate and graduate students in lab sessions, explaining course concepts, coding assignments, and practical applications.
- ❖ Mentored students in academic writing, designing and executing semester projects and research papers.
- ❖ Reviewed and provided feedback on student research projects, conference submissions, and journal papers.

Courses: Data Science, Artificial Intelligence, Machine Learning

RESEARCH EXPERIENCE & PUBLICATIONS | [Google Scholar](#)

Undergraduate Research Work September 2022 – September 2023

Thesis Title: Image Processing & Deep Learning Based Road Object Detection System for Safe Transportation.

Supervisor: Dr. Md. Nawab Yousuf Ali

- ❖ DhakaRoadNet: An Offline Edge AI Framework for Road-Object Detection in Dhaka Traffic Using YOLOv8n and TensorFlow Lite. *Manuscript submitted to COMPAS 26; First Author*
- ❖ Impact of AI in Healthcare Services: Analysis Using Medical Synthetic Data.
- ❖ Image Processing & Deep Learning Based Road Object Detection System for Safe Transportation.
- ❖ Liver Tumor Detection Using CLAHE Histogram Equalization and U-Net Model From 3D CT Scan Images.
- ❖ Comprehensive Dataset of Annotated Rice Panicle Image from Bangladesh

TECHNICAL SKILLS

- ❖ **Programming Languages:** Python, Kotlin, Java, C++, SQL
- ❖ **Machine Learning & Data Science:** Data preprocessing, exploratory data analysis, feature engineering, supervised learning, model training, evaluation, hyperparameter tuning, and performance optimization
- ❖ **Deep Learning:** Artificial neural networks, backpropagation, activation functions, regularization, batching, early stopping, checkpointing, and learning-rate tuning
- ❖ **Computer Vision:** Image preprocessing, dataset annotation, object detection, YOLO-based model training and evaluation, real-world testing, and edge deployment
- ❖ **Natural Language Processing:** Text preprocessing, tokenization, feature extraction, text classification, embeddings, sequence modeling, and transformer-based concepts
- ❖ **Edge AI & Deployment:** TensorFlow Lite and Android-based real-device deployment
- ❖ **Software Engineering:** Object-oriented programming, data structures and algorithms, design patterns, clean architecture, REST API integration, concurrency, multithreading, and performance optimization
- ❖ **Android Development:** Android application development using Kotlin/Java
- ❖ **Tools & Platforms:** Git, GitHub, Android Studio, Jupyter Notebook, Google Colab, Firebase, LaTeX

PROGRAMMING COMPETITION EXPERIENCE

- ❖ Solved 1,500+ programming problems across online judges, including [Codeforces](#), [LeetCode](#), and [HackerRank](#), developing strong foundations in algorithms, data structures, graph theory, number theory, and dynamic programming.
- ❖ 1st Runner-Up, Intra-University Programming Contest, Spring 2021.
- ❖ 2nd Runner-Up, EWU Victory Day Programming Battle, 2021.
- ❖ 2nd Runner-Up, Intra-University Programming Contest, 2022.
- ❖ 2nd Runner-Up, EWU Intra Kick-Off Contest, Fall 2022.
- ❖ Honorable Mention, Inter-University Programming Contest, BUET, 2022.
- ❖ Top 15%, ICPC Asia Dhaka Regional Online Preliminary Contest, 2022.

PROJECTS

❖ DhakaRoadNet – EdgeAI | [GitHub Link](#)

DhakaRoadNet is an end-to-end edge AI road object detection system focused on urban traffic environments in Dhaka, Bangladesh. The project covers the full pipeline from custom dataset creation to model training, evaluation, TensorFlow Lite export, and Android app deployment.

Tools & Tech: Python, PyTorch, YOLOv8n, TensorFlow Lite, Android SDK (Kotlin), CameraX

❖ Advanced Regression Analysis Using Classical ML and Ensemble Learning | [GitHub Link](#)

This project is a complete end-to-end machine learning pipeline built on the Ames Housing dataset, demonstrating real-world regression modeling, feature engineering, and advanced ensemble techniques.

Tools & Tech: ML, Regression Analysis, Feature Engineering, Ensemble Learning, Model Evaluation

❖ ScholarRAG - Local Citation-Grounded Research Paper Q&A Chatbot | [GitHub Link](#)

ScholarRAG is a fully local Retrieval-Augmented Generation (RAG) chatbot for question answering over English research papers. Users can upload PDFs, index their content, and ask questions through a Gradio interface. The system retrieves relevant passages and generates evidence-grounded answers with paper and page-number citations.

Tools & Tech: Python, NLP, RAG, LLMs, ChromaDB, Ollama, Sentence Transformers, Gradio